

TOTAL PANEL 3 LOAD						24V	83.96	Total	179		
Lighting Loads	Circuit	Feet	Watts	Amps	Recommended Trans Breaker						
Circuit 1	P3-EXT-1	35	2,747.36	198.87					93.75		
Circuit 2	P3-EXT-2	15	160.01	6.67	HDR-240-24 (10A)	10A	6.67	Shelly RGBWW Pro	Dimmer		
Circuit 3	P3-EXT-3	45	205.73	8.67	HDR-240-24 (10A)	10.7 A	8.67	Shelly RGBWW Pro	Dimmer		
Circuit 4	P3-EXT-4	65	297.17	12.38	HDR-320-24 (13.3A)	15.5 A	12.38	Shelly RGBWW Pro	Dimmer		
Circuit 5	P3-EXT-5	40	182.87	7.62	HDR-240-24 (10A)	9.5 A	7.62	Shelly RGBWW Pro	Dimmer		
Circuit 6	P3-LVC-1	Living	360.00	15.00	HDR-480-24 (20A)	20 A	15.00	Shelly RGBWW Pro	Dimmer		
Circuit 7	P3-LVC-2	Shower	120.00	5.00	HDR-240-24 (10A)	7A	5.00	Shelly RGBWW Pro	Dimmer		
Circuit 8	P3-LVC-3	Bath	168.00	7.00	HDR-320-24 (13.3A)	10A	7.00	Shelly RGBWW Pro	Dimmer		
Circuit 9	P3-LVC-4	Master	312.00	13.00	HDR-480-24 (20A)	15 A	13.00	Shelly RGBWW Pro	Dimmer		
Circuit 10	P3-LVM-1	Laundry	18.00	0.75	HDR-480-24 (20A)	16 A		Shelly PM-4-1.1		3.00	
	P3-LVM-2	Mechanical	18.00	0.75	Combined 24V-10			Shelly PM-4-1.2			
	P3-LVM-3	WC-1	18.00	0.75	Combined 24V-10			Shelly PM-4-1.3			
	P3-LVM-4	WC-2	18.00	0.75	Combined 24V-10			Shelly PM-4-1.4			
	P3-LVD-1	Closet	288.00	12.00	Combined 24V-10						
Circuit 11	P3-120-1	Entry	75.00	3.13	HDR-480-24 (20A)	20 A	12.65	Shelly PM-4-2.1			
	P3-120-2	Foyer	75.00	3.13	Combined 24V-11			Shelly PM-4-2.2			
	P3-120-3		75.00	3.13	Combined 24V-11			Shelly PM-4-2.3			
	P3-F1-1			1.00	Combined 120V-11			Shelly PM-4-2.4			
	P3-F-1			1.00	Combined 120V-11			Shelly 2 PM Pro-1.1	Dimmer		
	P3-F1-2			1.00	Combined 120V-11			Shelly PM-1			
	P3-F-2			1.00	Combined 120V-11			Shelly 2 PM Pro-1.2	Dimmer		
Wall Loads											
Circuit 12	P1-120-1 (120V)			20	120V Circuit	40 A	16	Shelly PM-3-3.1			
	P2-120-1 (120V)			20	120V Circuit	20 A	16	Shelly PM-3-3.2			
	P3-120-1 (120V)			20	120V Circuit	20 A	16	Shelly PM-3-3.3			
							16	Shelly PM-3-3.4			
	220-1 (220V)			15	220V Circuit	20 A	16	Shelly PM-2-2.1			
	220-2 (220V)			15	220V Circuit	20 A	16	Shelly PM-2-2.2			
Blinds											
Circuit 11	Entry			57.6 0.48	120V Circuit		0.48	Shelly Dual Cover 1.1			
	Entry 2			57.6 0.48	120V Circuit		0.48	Shelly Dual Cover 1.2			
	West Glass Upper 1			57.6 0.48	120V Circuit		0.48	Shelly Dual Cover 2.1			
	West Glass Upper 2			57.6 0.48	120V Circuit		0.48	Shelly Dual Cover 2.2			
	Glass South			57.6 0.48	120V Circuit		0.48	Shelly Dual Cover 3.1			

TOTAL PANEL 1 LOAD											
Lighting Loads	Load	Feet/Location	Watts	Load A	Transformer	Breaker	NCL A				
Total			3,228.90	127.34			127.34				
Circuit 1	P1-EXT-1		160.02	6.67	HDR-240-24 (10A)	7A	6.67	Shelly RGBWW Pro	Dimmer		
Circuit 2	P1-EXT-2		297.18	12.38	HDR-480-24 (20A)	15 A	12.38	Shelly RGBWW Pro	Dimmer		
Circuit 3	P1-EXT-3		228.80	9.53	HDR-320-24 (13.3A)	10A	9.53	Shelly RGBWW Pro	Dimmer		
Circuit 4	P1-EXT-4		388.62	16.19	HDR-480-24 (20A)	20 A	16.19	Shelly RGBWW Pro	Dimmer		
Circuit 5	P1-EXT-5		274.32	11.43	HDR-320-24 (13.3A)	15 A	11.43	Shelly RGBWW Pro	Dimmer		
Circuit 6	P1-EXT-6		320.04	13.34	HDR-480-24 (20A)	15 A	13.33	Shelly RGBWW Pro	Dimmer		
Circuit 7	P1-EXT-7		274.32	11.43	HDR-320-24 (13.3A)	15 A	11.43	Shelly RGBWW Pro	Dimmer		
Circuit 8	P1-LVC-1	Kitchen Cooktop	168.00	7.00	HDR-320-24 (13.3A)	15 A	7	Shelly RGBWW Pro	Dimmer		
Circuit 1	P1-LVC-2	Kitchen dish Sink	72.00	3.00	HDR-150-24 (5A)	7A	3	Shelly RGBWW Pro	Dimmer		
Circuit 9	P1-LVC-3	Office	360.00	15.00	HDR-480-24 (20A)	15 A	15	Shelly RGBWW Pro	Dimmer		
Circuit 10	P1-LVM-1	Garge	36.00	1.50	HDR-320-24 (20A)	15 A	12	Shelly 1 PM			
	P1-LVM-2	Systems Garage	18.00	0.75	Combined 24V-10			Shelly 1 PM			
	P1-LVM-3	Back Stair	36.00	1.50	Combined 24V-10			tx	Dimmer		
	P1-LVM-4	Mechanical	18.00	0.75	Combined 24V-10			Shelly 1 PM			
	P1-LVM-5	Storage	18.00	0.75	Combined 24V-10			Shelly 1 PM			
	P1-LVM-6	Pantry	54.00	2.25	Combined 24V-10			Shelly 1 PM			
	P1-LVM-7	Office Closet	36.00	1.50	Combined 24V-10			Shelly 1 PM			
	P1-LB-8	Office Bath	72.00	3.00	Combined 24V-10			Shelly 2 PM Pro-1.1	Dimmer		
Circuit 11	P1-L120-1	Powder	75.00	3.13	HDR-480-24 (20A)	15 A	9.375	Shelly 1 PM			
	P1-L120-2	Island	75.00	3.13	Combined 120V-11			Shelly 2 PM Pro-1.1	Dimmer		
	P1-L120-3	Dining	75.00	3.13	Combined 120V-11			Shelly 2 PM Pro-1.1	Dimmer		
Wall Loads											
Total				231.44			294.44				
Circuit 12	P1-120-1 (120V)		16.00		uit	40 A GFCI		Shelly PM-4-1.1			
	P1-120-2 (120V)		16.00				16	Shelly PM-4-1.2			
	P1-120-3 (120V)		16.00		120V Circuit		16	Shelly PM-4-1.3			
	P1-120-4 (120V)		16.00		120V Circuit		16	Shelly PM-4-1.4			
	P1-120-5 (120V)		16.00		120V Circuit	40 A GFCI	16	Shelly PM-4-2.1			
	P1-120-6 (120V)						16	Shelly PM-4-2.2			
	P1-120-7 (120V)						16	Shelly PM-4-2.3			
	P1-120-8 (120V)						16	Shelly PM-4-2.4			
	P1-120-9 (120V)					15	15	Shelly 1 PM			
Circuit 13	220-1 (220V)			16.00	220V Circuit	40 A GFCI	16	Shelly PM-4-3.1			
	220-2 (220V)			16.00	220V Circuit		16	Shelly PM-4-3.2			
	220-3 (220V)			16.00	220V Circuit		16	Shelly PM-4-3.3			
	220-4 (220V)			16.00	220V Circuit		16	Shelly PM-4-3.4			
Circuit 14	220-5 (220V)			16.00	220V Circuit	15A	16	Shelly 1 PM			
Circuit 15	220-6 (220V)			30.00	220V Circuit	30 A	30				
Circuit 16	220-7 (220V)			40	220V Circuit	40 A	40				
Blinds											
Circuit 3	Lower West 1			57.6 0.48	120V Circuit		1.44	Shelly Dual Cover 1.1			
	Lower West 2			115.2 0.96	Combined 120V-3			Shelly Dual Cover 1.2			

TOTAL PANEL 4 LOAD											
Lighting Loads	Circuit	Feet	Watts	Amps	Recommended Trans Breaker						
Circuit 1	P4-LVS-1	40	182.80	7.62	HDR-240-24 (10A)	10 A	7.62	Shelly 2 PM Pro-1.1	Dimmer		
Circuit 2	P4-LVS-2	40	182.80	7.62	HDR-240-24 (10A)	10 A	7.62	Shelly 2 PM Pro-1.2	Dimmer		
Circuit 3	P4-LVS-3	65	297.00	12.38	HDR-320-24 (13.3A)	15 A	12.38	Shelly PM-1			
Circuit 4	P4-LVC-1	Overhead Cans	360.00	15.00	HDR-480-24 (20A)	20 A	15.00	Shelly RGBWW Pro	Dimmer		
Circuit 5	P4-LVM-1	Mechanical	18.00	0.75	HDR-150-24 (6.3A)	3.25	2.50	Shelly PM-4-1.1			
	P4-LVM-2	Bath	24.00	1.00	Combined 24V-5			Shelly PM-4-1.2			
	P4-LVM-3	Escape 1	18.00	0.75	Combined 24V-6			Shelly PM-4-1.3			
	P4-LVM-4	Escape 2	18.00	0.75	Combined 24V-7			Shelly PM-4-1.4			
Wall Loads											
Circuit 6	P4-1			20.00	120V Circuit	15 A	16	Shelly PM-2-2.1			
Circuit 7	P4-2			20.00	120V Circuit	15 A	16	Shelly PM-2-2.2			
Circuit 8	220-1 (220V)			16.00	220V Circuit	15 A	16	Shelly PM-2-2.1			
Circuit 9	220-1 (220V)			16.00	220V Circuit	15 A	16	Shelly PM-2-2.2			
Circuit 10	220-2 (220V)			30.00	220V Circuit	30 A	30				

TOTAL PANEL 2 LOAD			4 071.02	161.23	Total		176.55		
Lighting Loads	Load	Feet/Location	Watts	Continuous A	Transformer	Breaker	NCL A		
Total			2,035.51	80.61			88.27		
Circuit 1	P2-EXT1	20	91.35	3.81	HDR-240-24 (10A)	5A	3.81	ShellyY RGBWW Pro	Dimmer
	P2-EXT2	35	159.86	6.66	HDR-240-24 (10A)	8A	6.66	ShellyY RGBWW Pro	Dimmer
Circuit 2	P2-EXT3	16	73.08	3.05	HDR-240-24 (10A)	5A	3.04	ShellyY RGBWW Pro	Dimmer
	P2-EXT4	40	182.70	7.61	HDR-240-24 (10A)	10A	7.61	ShellyY RGBWW Pro	Dimmer
Circuit 3	P2-EXT5	70	319.72	13.32	HDR-320-24 (13.3A)	15 A	13.32	ShellyY RGBWW Pro	Dimmer
Circuit 4	P2-LVC-1		216.00	9.00	HDR-320-24 (13.3A)	12A	9.00	ShellyY RGBWW Pro	Dimmer
Circuit 5	P2-LVC-2		240.00	10.00	HDR-320-24 (13.3A)	15 A	10.00	ShellyY RGBWW Pro	Dimmer
Circuit 6	P2-LVC-3		240.00	10.00	HDR-320-24 (13.3A)	16 A	10.00	ShellyY RGBWW Pro	Dimmer
Circuit 7	P2-LVM-1	Mechanical	18.00	0.75	HDR-480-24 (20A)	40	17.17	Shelly PM-4-1.1	
	P2-LVM-2	Closet	48.00	2.00	Combined 24V-7			Shelly PM-4-1.2	
	P2-LVM-4	Garage	36.00	1.50	Combined 24V-7			Shelly PM-4-1.3	
	P2-LVM-5	Garage	126.00	5.25	Combined 24V-7			Shelly PM-4-1.4	
Circuit 8	P2-LVM-3	Bath	72	3.00	HDR-150-24	10A	7.67	Shelly 2 PM Pro-1.1	Dimmer
	P2-FL-120-1		40.00	1.67	120V Circuit			Shelly 2 PM Pro-1.1	Dimmer
	P2-F-120-1			1.00	120V Circuit			Shelly PM-1	
	P2-FL-120-2			1.00	120V Circuit			Shelly 2 PM Pro-1.1	Dimmer
	P2-F-120-2			1.00	120V Circuit			Shelly PM-1	
Wall Loads									
				316.4			183.44		
Circuit 8	P1-120-1			20	120V Circuit			Shelly PM-4-2.4	
	P1-120-2			20	120V Circuit	40 A GFCI	16	Shelly PM-4-3.1	
	P1-120-3			20	120V Circuit		16	Shelly PM-4-3.2	
	P1-120-4			20	120V Circuit		16	Shelly PM-4-3.3	
	P1-120-5			20	120V Circuit		16	Shelly PM-4-3.4	
Circuit 9	P1-120-6			20	120V Circuit		15	Shelly PM-1	
	220-1 (220V)			20	220V Circuit	40A	16	Shelly PM-4.4.1	
	220-2 (220V)			20	220V Circuit		16	Shelly PM-4.4.2	
Circuit 10	220-3 (220V)			20	220V Circuit			Shelly PM-4.4.3	
								Shelly PM-4.4.4	
Circuit 11	220-4 (220V)			30	220V Circuit	30A	30		
Circuit 12	220-4 (220V)				40 220V Circuit	40A	40		
Blinds									
Circuit 4	Glass East		57.6	0.48	120V Circuit		0.96	Shelly Dual Cover 1.1	
	West Glass 2-1		57.6	0.48	120V Circuit			Shelly Dual Cover 1.2	
	West Glass 2-2		57.6	0.48	120V Circuit		0.48	Shelly Dual Cover 2.1	